Students: *David Hidalgo Fàbregas, Michael George Nabih Lotfy Eskander, Miriam Aguiló Iturralde, Pablo Longán Gasol, João Constantino Muianga.*

Supervised Learning Methods

Classifying Breast Cancer: A Comparative Analysis

List of libraries used: *time, pandas, numpy, matplotlib, seaborn, scikit-learn*

Index

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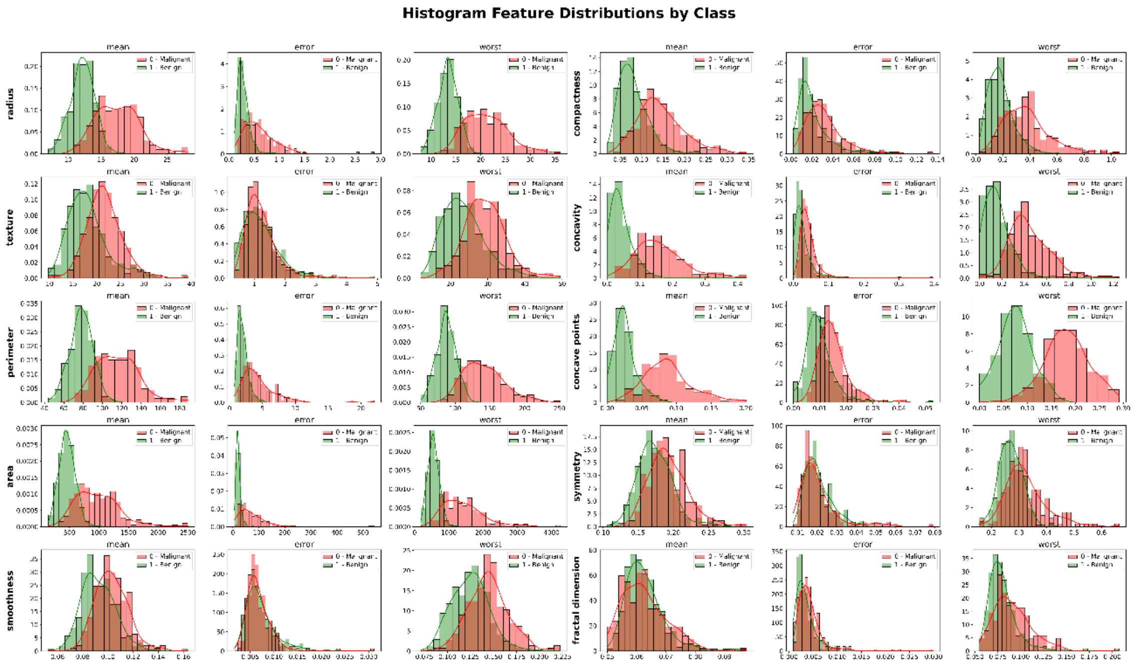
1. Introduction and Methodology

1.1. Dataset

This study uses the Breast Cancer Wisconsin (Diagnostic) Dataset, which comprises 569 records derived from digitized fine needle aspirate (FNA) images. The dataset includes 30 numerical predictors (float type) that describe the morphological characteristics of cell nuclei. These predictors are organized into ten fundamental measurements, such as radius, texture, and area, which are calculated using the mean, standard error, and 'worst' value.

Exploratory analysis confirms the high quality of the data, revealing an absence of missing values. The target variable is binary, classifying diagnoses as malignant (0) or benign (1), but it is slightly imbalanced (63% of the target are benign cases).

Distributions



Several attributes show visual distinction between malignant (red) and benign (green) tumors. radius, perimeter, area, and concave points (especially in their *mean* and *worst* metrics), the benign class is clustered at lower values while the malignant class shows higher values.

There are features where the red and green distributions overlap almost entirely, what makes them not strong differentiators for the target variable. This overlap is particularly noticeable in fractal dimension (*mean* and *error*), smoothness (*error*), and symmetry (*error*).

The benign class (green) tends to have narrower distributions with sharper peaks, indicating lower variance. The malignant class (red) displays greater dispersion, and many of its distributions are right-skewed (featuring long tails). This pattern is evident across almost all *error* metrics (e.g., area error, perimeter error).

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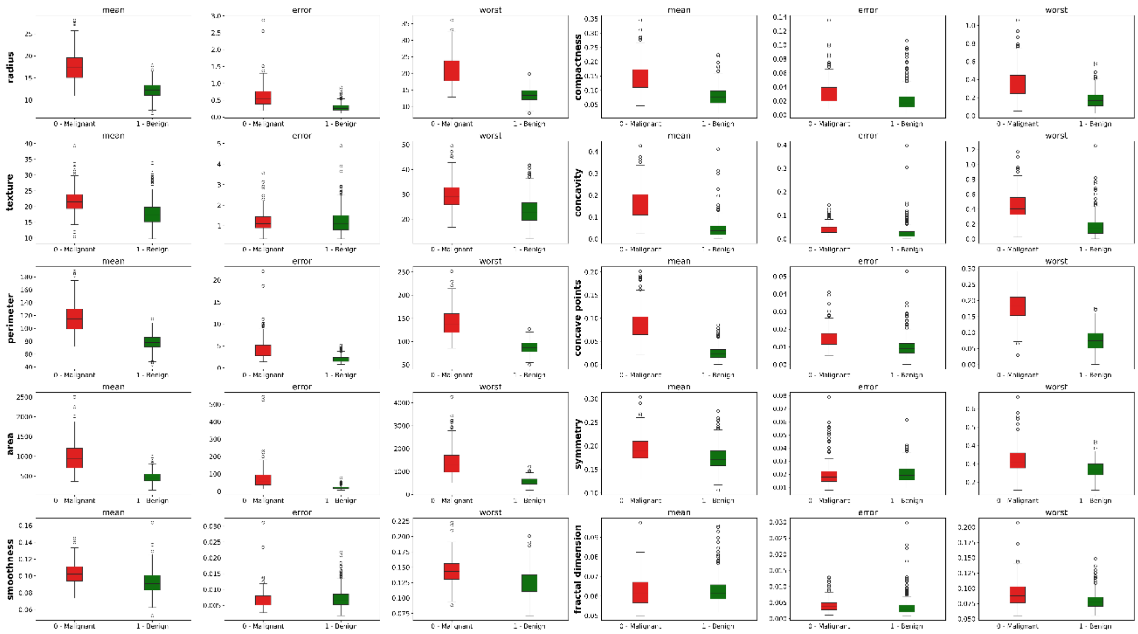
Boxplot Feature Distributions by Class



Medians and Interquartile Range (IQR): Boxplots show that malignant tumors present higher median values across *size* and *severity* metrics (mean radius, mean perimeter, mean area and mean concavity). The interquartile ranges, i.e., the boxes themselves representing the middle 50% of the data, are also generally wider for the malignant class, confirming the higher variance and structural inconsistency.

Right-skewed: Boxplots strongly emphasize the right-skewed nature of the malignant class. Lots of features display extremely long upper whiskers and numerous high-value outliers. This reflects the uncontrolled growth typical of cancerous cells in comparison with the benign class, which remains much more tightly clustered at lower values.

Confirmation of Separability: Almost all *mean* and *worst* feature metrics such as worst perimeter, worst area, or worst concave points present boxes that separate completely, meaning that they are strong discriminators. Almost all *error* feature metrics such as smoothness error, symmetry error, or even the mean fractal dimension show overlapping boxes and medians that are similar, confirming they offer low predictive power on their own.

Scale Disparity: There is a massive difference in magnitudes across different features. This evidence justifies the need to apply StandardScaler in the preprocessing stage to prevent features with large absolute values from dominating distance-based algorithms.

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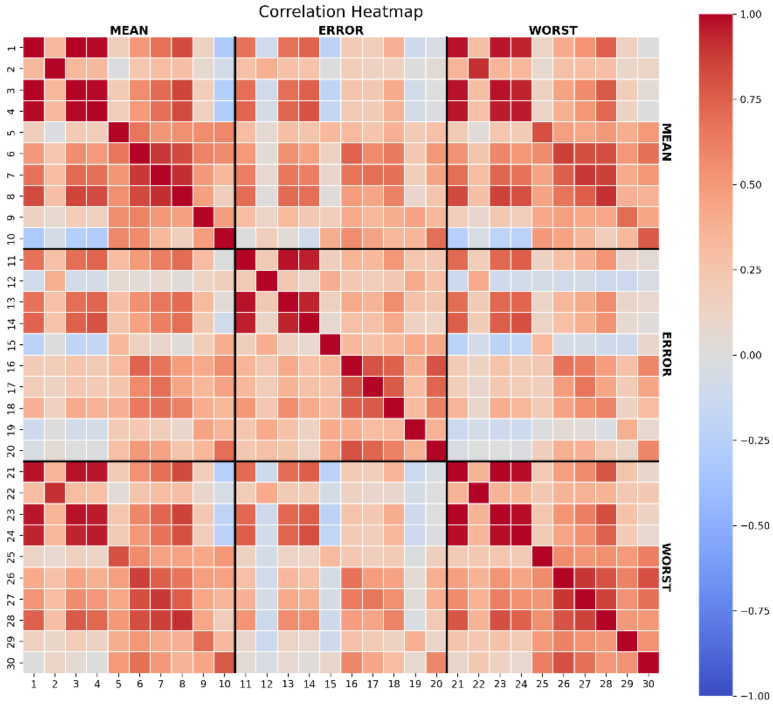
Top correlated features



ID	Feature Legend
1	mean radius
2	mean texture
3	mean perimeter
4	mean area
5	mean smoothness
6	mean compactness
7	mean concavity
8	mean concave points
9	mean symmetry
10	mean fractal dimension
11	radius error
12	texture error
13	perimeter error
14	area error
15	smoothness error
16	compactness error
17	concavity error
18	concave points error
19	symmetry error
20	fractal dimension error
21	worst radius
22	worst texture
23	worst perimeter
24	worst area
25	worst smoothness
26	worst compactness
27	worst concavity
28	worst concave points
29	worst symmetry
30	worst fractal dimension

Size Metrics: The strongest correlations exist between radius, perimeter, and area across all three of their metrics (*mean*, *error*, and *worst*). This makes sense, as perimeter and area are functions of the radius.

Shape Severity: Another cluster of high positive correlation is found between compactness, concavity, and concave points. This makes sense because a cell nucleus with a highly irregular, concave contour will have more concave points and a higher compactness score.

Mean vs Worst Redundancy: There is a strong correlation between the *mean* values and the *worst* values for almost every single attribute.

1.2. Preprocessing

Features data were stratigraphically split by the target column, preserving the class variance.

After that, Standard Scaler were applied to delete scale bias. Standard Scaler was chosen over MinMax Scaler because it is better suited for the specific algorithms evaluated in this assignment. Centering the features at a mean of 0 helps models to learn faster and more stably. Furthermore, setting the standard deviation to 1 ensures that all variables contribute equally to the models. This prevents features with naturally larger numbers from unfairly dominating distance calculations in KNN or biasing the regularization processes in Logistic Regression and SVM. Standard Scaler handles clinical outliers much better, whereas MinMax Scaler would compress most of our normal data into a tiny range if extreme values were present.

Then, two complementary filtering approaches were applied.

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Near-zero variance filtering

Before applying Standard Scaler for modeling, we evaluated the original variance of each feature to identify variables with near-zero variance. It was crucial to perform this step before scaling, since Standard Scaler forces all features to have a variance of 1, which would have masked any low variability.

However, calculating the variance on the raw data would have introduced a scale bias. That is why we temporarily applied a MinMax Scaler to map all features to a common range of $[0, 1]$ while preserving their relative distributions. Finally, features with a variance less than 0.01 were suppressed, which correspond to the ones that present less than a 10% deviation from the mean.

Correlation filtering

After applying the Standard Scaler, we evaluated the correlation between features to identify and mitigate severe multicollinearity, which is particularly prevalent in this dataset due to mathematically linked geometric features such as tumor radius, perimeter, and area. We established a correlation threshold of 0.9 because variables exceeding this limit share over 81% of their variance, providing essentially duplicate information. By discarding redundant features, we reduce unnecessary dimensionality and stabilize the learned weights (crucial in Logistic Regression), thereby simplifying the overall architecture and preventing overfitting without sacrificing meaningful predictive power.

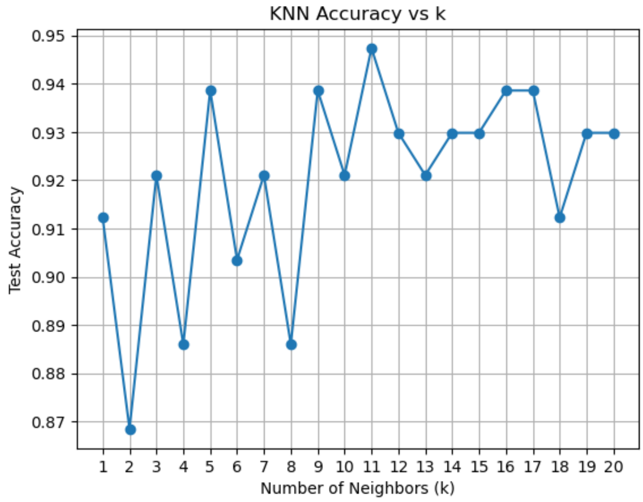
After both filters, 13 features were suppressed.

2. Algorithm Comparison

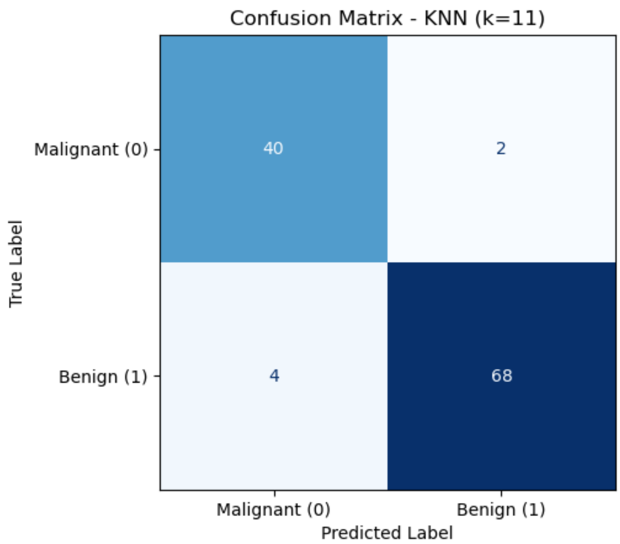
2.1. Model Adjustment

k-Nearest Neighbors

The KNN model was evaluated using values of *k* ranging from 1 to 20. The results showed that the highest accuracy was achieved at *k = 11*, with a test accuracy of 94.74%. Smaller values of *k* captured local data patterns more effectively, while larger values reduced performance due to overly smooth decision boundaries.



The classification report and confusion matrix demonstrated strong predictive performance, with high precision, recall, and relatively few misclassifications. However, some malignant cases were incorrectly classified as benign, which is an important limitation in medical diagnosis.



Overall, KNN proved to be an effective and simple classification method, although its performance depends heavily on the choice of *k* and proper feature scaling.

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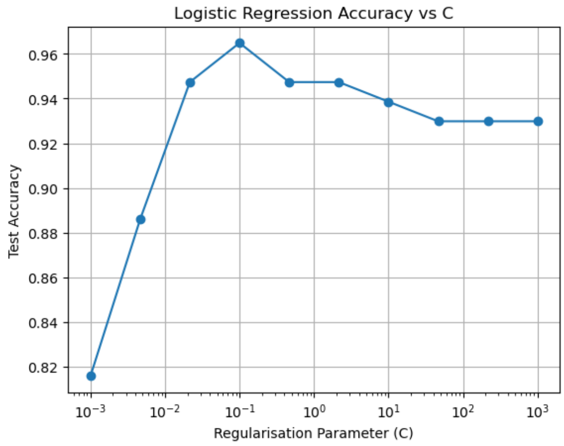
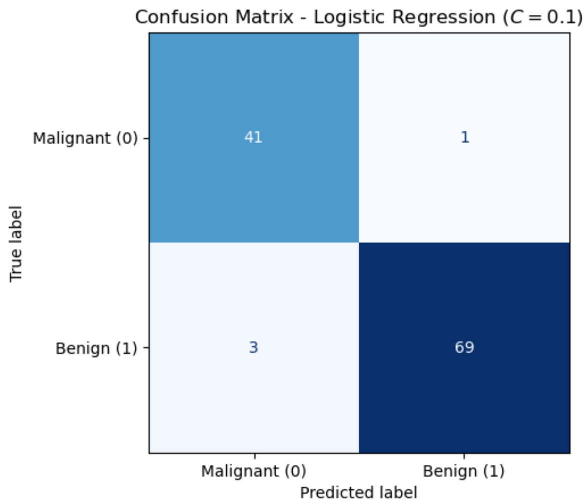
Logistic Regression

The Logistic Regression model achieved excellent predictive performance on the breast cancer dataset. After testing multiple regularization strengths, the optimal model was obtained at $C = 0.1$, achieving an accuracy of approximately 96%.



The classification report showed very high precision, recall, and F1-scores for both classes, while the confusion matrix confirmed very few classification errors. Coefficient analysis also showed that features related to tumor size, texture, and shape were the most influential predictors.



Compared to KNN, Logistic Regression achieved higher accuracy and better overall generalization. Its strong interpretability and reliable performance make it highly suitable for medical classification tasks such as breast cancer diagnosis.

Support Vector Machine

The SVM task began by comparing three kernel types Linear, RBF, and Polynomial trained on the 17-feature StandardScaled dataset with identical default settings (C=1, gamma='scale'). This ensured the comparison isolated the effect of the kernel itself rather than any tuning advantage.

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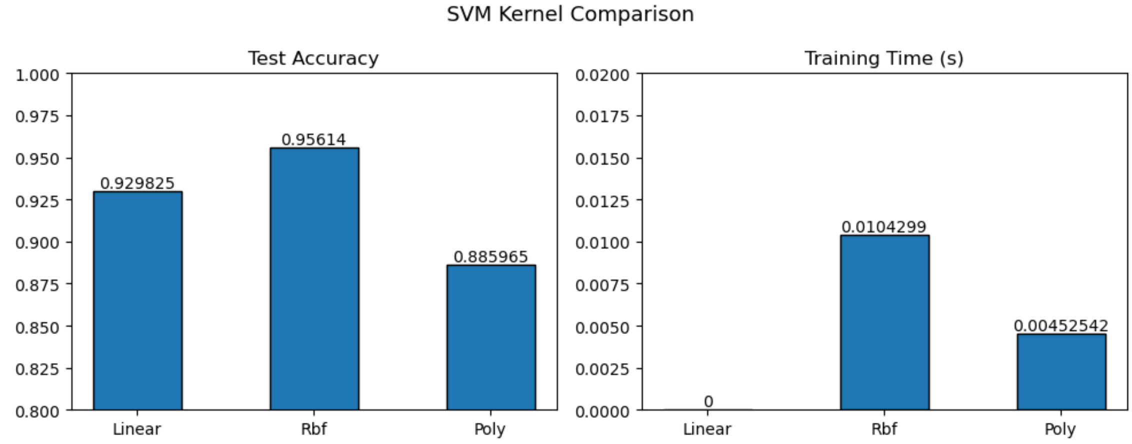
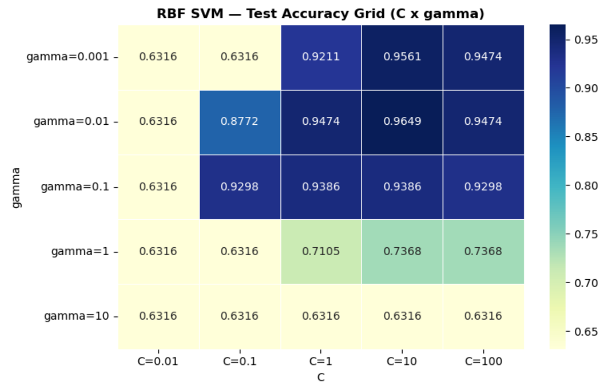
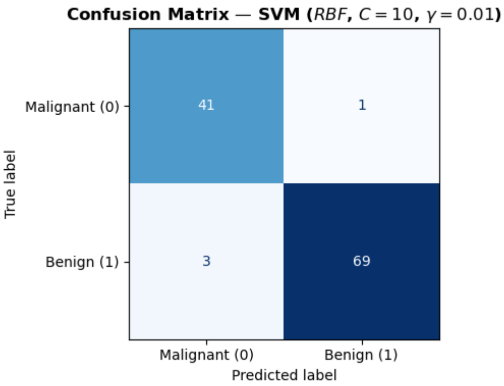


Among the three kernels, RBF achieved the highest test accuracy at 95.61%, followed by Linear at 92.98%, and Polynomial at 88.60%. The polynomial kernel's weaker performance suggests that high-degree feature interactions do not add meaningful signal beyond what RBF already captures. Based on these results, the RBF kernel was selected for further hyperparameter tuning.

A *C vs. gamma* grid search was then conducted on the RBF kernel, sweeping C across [0.01, 0.1, 1, 10, 100] and gamma across [0.001, 0.01, 0.1, 1, 10]. The optimal combination of C=10 and gamma=0.01 pushed the final model's test accuracy to 96.49%, with a macro F1-score of 0.96. Notably, the model achieved a malignant recall of 0.98, meaning it correctly flagged 98% of true malignant cases a critical property in a clinical screening context where false negatives carry the highest cost.





Neural Network

The neural network was implemented using scikit-learn's MLPClassifier. The adjustment process was carried out in three sequential steps.

Activation Function Selection: Three activation functions were compared on a fixed architecture of (16, 8) neurons: relu, tanh, and logistic. relu was selected because it achieved the steepest loss drop in the early epochs and stabilised fastest, indicating more efficient gradient propagation for this dataset.

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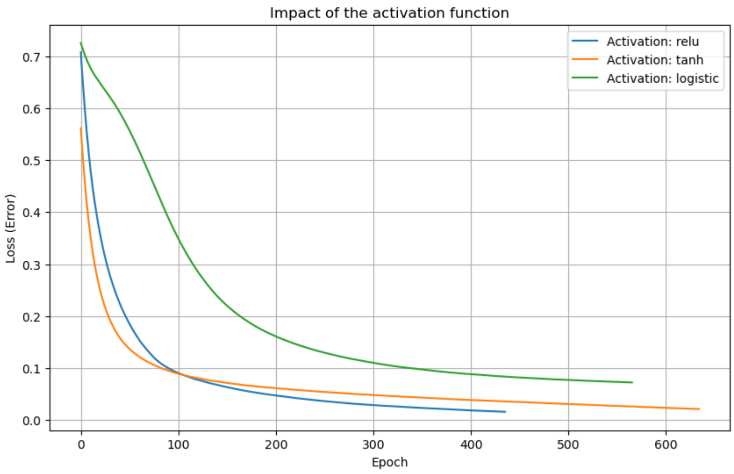
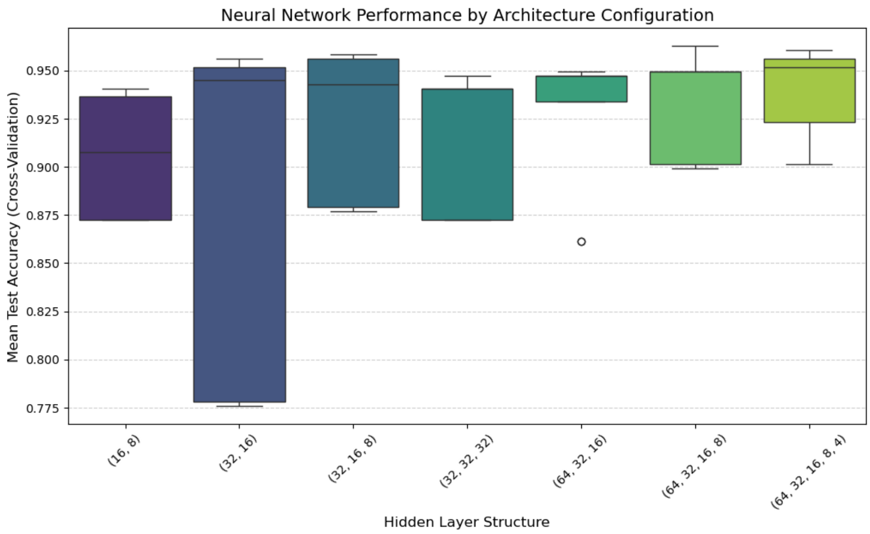


Architecture and Hyperparameter: Search a GridSearchCV with 3-fold cross-validation was run over the following search space:

Parameter	Values tested
Hidden_layer sizes	(16,8), (32,16), (32,16,8), (32,32,32), (64,32,16), (64,32,16,8), (64,32,16,8,4)
Alpha (L2 regularization)	0.0001, 0.01, 0.1
Learning rate	0.001, 0.01, 0.1

The best configuration found was: architecture (64, 32, 16, 8), alpha = 0.0001, learning rate = 0.1. The architecture comparison boxplot confirmed that (64, 32, 16, 8) reached the highest cross-validated accuracy among all candidates.



Final Parameter Confirmation via Heatmap: With the best architecture fixed, a heatmap of mean cross-validated accuracy across all (alpha, learning rate) combinations was used to visually confirm the optimal pair (0.0001, 0.1), ensuring the selection was not an artefact of architecture alone.

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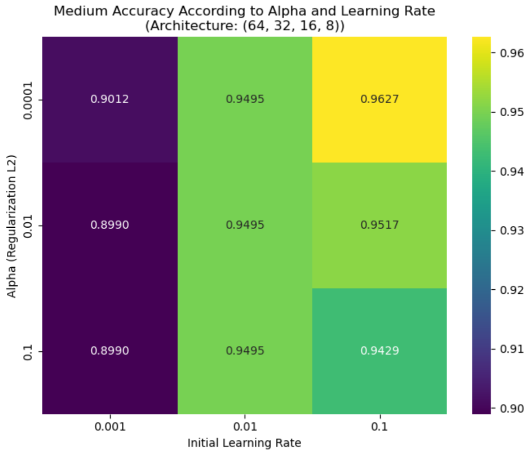
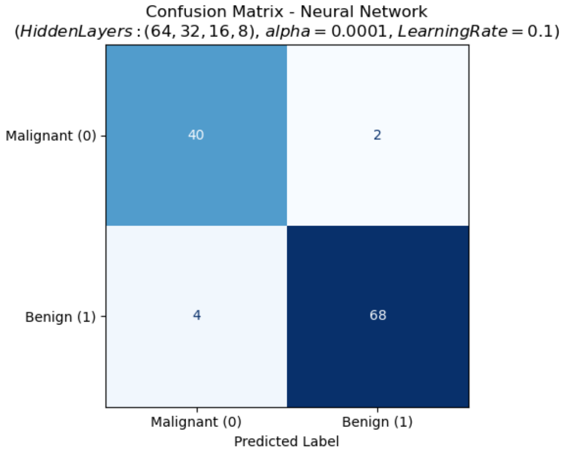
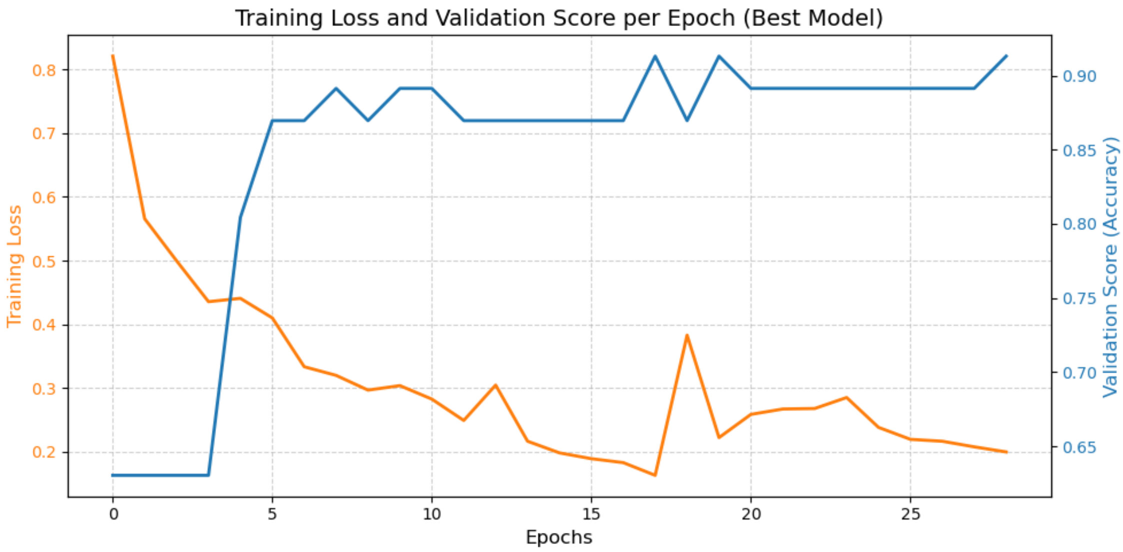




The learning curve for the optimal Neural Network configuration illustrates the model's convergence behavior across epochs. During the initial phase, network exhibits rapid learning, characterized by a steep decline in training loss and a corresponding surge in validation accuracy. While the training process displays some volatility, the model quickly stabilizes. This demonstrates that the network effectively generalized the patterns from the training set without suffering from severe overfitting.



2.2. Assumptions, strengths and limitations

The following table provides a qualitative comparison of the machine learning models considered in this study. For each algorithm, the main assumptions, strengths, and limitations are summarized to better understand their theoretical characteristics and practical suitability for the classification task.

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Model	Assumptions	Strengths	Limitations
KNN	Similar observations are located close to each other in the feature space.	Simple and effective for classification tasks with well-structured data.	Computationally expensive for large datasets because predictions require comparison with all training samples. Highly sensitive to the scale of features, requiring carefull preprocessing.
Logistic Regression	Assumes a linear relationship between features and the target variable. Assumes little to no multicollinearity among independent variables.	Highly interpretable and computationally efficient.	Performance may decrease when the relationship between variables is highly nonlinear.
SVM	Classes can be separated by a hyperplane in the feature space, either directly or through a kernel transformation.	Highly effective in high-dimensional spaces and can handle non-linear boundaries well through the kernel trick. The decision boundary relies only on support vectors, making the model memory efficient.	Sensitive to hyperparameter choices and scale poorly to very large datasets, as training time grows quadratically with sample size.
Neural Network	Complex relationships can be learned through layered representations.	Captures non-linear patterns, is highly flexible, and has a strong predictive performance.	Requires more computational resources, is less interpretable and prone to overfitting without tuning. Typically requires very large datasets to generalize well and avoid overfitting.

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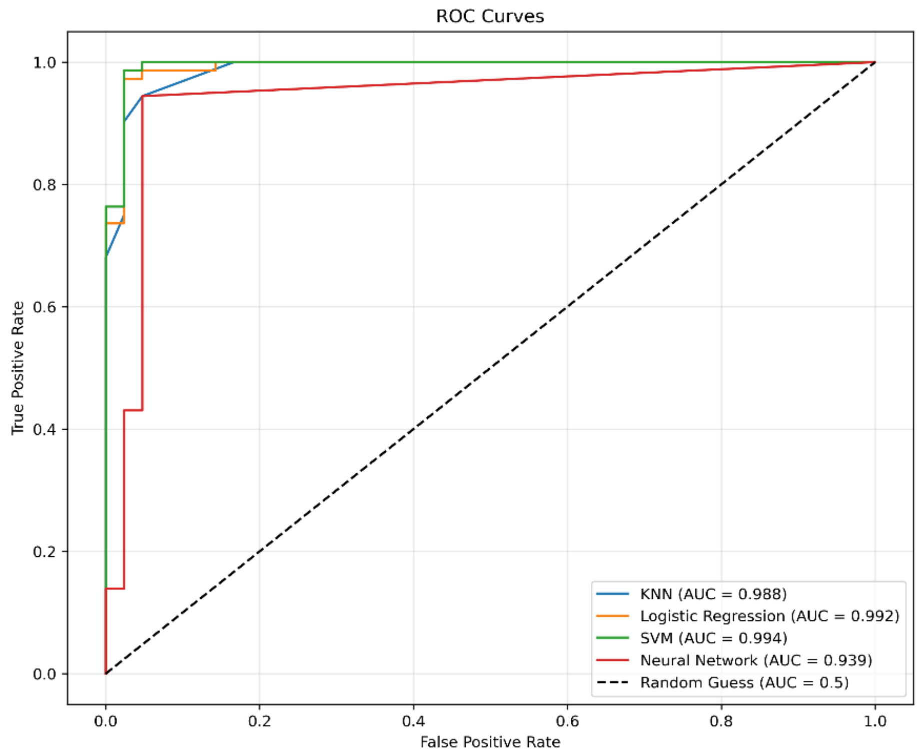
3. Evaluation

All models achieved strong predictive performance:



The analysis of the ROC curves reveals outstanding performance across most classifiers, suggesting that the geometric characteristics of the cell nuclei in the Wisconsin breast cancer dataset are highly separable. SVM and Logistic Regression models demonstrated the highest discriminative capacity, achieving the AUC values of 0.994 and 0.992, respectively. K-Nearest Neighbors also showed highly competitive performance with an AUC of 0.988. In contrast, the Neural Network exhibited the lowest performance (AUC = 0.939) and a more stepped curve, which could indicate slight overfitting due to the architecture's complexity.

The following table summarizes the performance of the machine learning models evaluated in this study. Each model is compared using key evaluation metrics, including accuracy, macro F1-score, AUC, and training time, to assess both predictive performance and computational efficiency.

Model	Accuracy	F1-Score (Macro)	AUC	Training Time (s)
KNN	0.9474	0.944	0.9881	0.002
Logistic Regression	0.9649	0.9627	0.9917	0.004
SVM (RBF)	0.9649	0.9627	0.994	0.0186
Neural Network	0.9474	0.944	0.9395	0.1871

Considering the clinical context of the problem, Logistic Regression is the most recommended model for deployment. It offers a predictive capability virtually identical to the best-performing model (SVM). Both Logistic Regression and SVM achieved the highest Accuracy and Macro F1-Score. However, Logistic Regression accomplished this

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with a significantly lower training time (0.004 seconds) compared to the SVM (0.0186 seconds) due to the significantly lower compute cost. Also, Logistic Regression has the critical advantage of being fully interpretable, which allows clinicians to directly understand the impact of individual features through coefficient weights.

3.1. Reflection

The performance and selection of the models in this analysis were influenced by the specific characteristics of the Breast Cancer Wisconsin dataset:

Dimensionality and Feature Correlation

With 30 real-valued features derived from 10 base measurements, the dataset presents a moderate dimensionality. The high correlation between features like radius, perimeter, and area benefits models like Logistic Regression and SVM, which can effectively handle linear dependencies through coefficient weighting or kernel transformations. In contrast, this redundancy can introduce noise for KNN, as the distance calculations become less meaningful in higher-dimensional space.

Sample and Normalization

The features present different scales. The mandatory use of Standard Scaler was critical. Distance-based models like KNN and SVM, as well as gradient-based Neural Networks, would have failed to converge or biased heavily toward large-magnitude features without this preprocessing step.

Sample Size and Complexity

The dataset contains 569 patient records, which is sufficient size for statistical models like Logistic Regression but proved challenging for the Neural Network. The stepped ROC curve and lower AUC suggest that the model's complexity slightly exceeded the information density of the training set, leading to a higher risk of overfitting compared to simpler, more robust estimators.

Class Balance and Clinical Metrics

The dataset features a mild, realistic imbalance (approximately 63% benign to 37% malignant). Artificially forcing a 50-50 class balance through oversampling or undersampling may have introduced synthetic noise or discard valuable patient data from an already limited sample of 569 records. Preserving the native distribution via a stratified split ensures the models learn realistic clinical probabilities rather than distorted once. Since clinical priority is minimizing false negatives (malignant cases missed), relying on metrics like the Macro F1-score (0.9627) and AUC (0.992) is highly effective. These metrics are inherently robust to mild imbalances and provide a fairer assessment of true clinical utility than raw accuracy.

3.2. Limitation

A limitation of this study is the reliance on a single train-test split. Given the relatively small sample size of 569 records, the results may be sensitive to the specific composition of the test set. Future iterations should implement k-fold cross-validation to ensure the reported metrics are stable and provide a more generalized estimate of model performance.

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